Quantova Inc

Quantova Bridge Guard Benchmark and Smoke Test

Quantova Inc benchmark report, October 2026

Release v0.3.0
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Summary

This report records the benchmark and smoke test results for Quantova Bridge Guard v0.3.0, the post quantum shield for bridges on Ethereum built by Quantova Inc. It measures the gas cost of every on chain operation for every supported LMS parameter set, the worst case cost of committee rotation against the Ethereum transaction gas cap, the performance of the signer kit, and a complete live run on a local Ethereum node.

Result	Value
Contract tests, including fuzzing, RFC 8554 vectors and Solidity and Python parity	55 of 55 passed
Signer kit RFC 8554 vector checks	15 of 15 passed
Signer kit safety checks	48 of 48 passed
Live smoke test steps on a local Ethereum node	12 of 12 passed
Worst case committee rotation against the EIP 7825 cap of 16,777,216 gas	12.44M gas, 26 percent headroom
Single message acceptance by inclusion proof in a batch of 256	7k gas of execution
Single packet delivery through LayerZero in a batch of 256	129k gas of execution

Every operation fits comfortably within the limits of Ethereum for every supported parameter set, and every functional and safety test passes. On these results Quantova Bridge Guard is ready for bridge teams to integrate as a post quantum shield, starting alongside their existing security as the integration guide describes.

Environment and Method

Item	Detail
Release	Quantova Bridge Guard v0.3.0
Machine	Apple M4, macOS 15.7.4
Toolchain	Foundry 1.7.1 with Anvil, solc 0.8.28 with the IR pipeline, Cancun EVM, optimizer runs 10000
Kit runtime	Python 3.14.7, standard library only
Continuous integration	Ubuntu with Python 3.10 and 3.12, every push

Contract figures are execution gas measured inside the EVM with the Foundry test runner, using real LMS signatures produced by the signer kit and real secp256k1 signatures. Execution gas excludes the 21,000 gas base cost and calldata cost of a transaction. Transaction figures from the smoke test include both. Every LMS figure is drawn from 16 signatures over distinct messages, because the Winternitz chain lengths, and therefore the cost, depend on the message.

LMS Signature Verification

Execution gas for one LMS verification through the SHA 256 precompile, by tree height and Winternitz parameter. The last column is the budget the guard reserves for that parameter set when it checks that a committee can always execute.

Parameters	Minimum	Mean	Maximum	Signature	Guard budget
H5 W1	225,938	231,068	236,594	8,684 bytes	291,500
H5 W2	151,835	157,881	165,661	4,460 bytes	236,500
H5 W4	184,549	197,507	219,105	2,348 bytes	361,500
H5 W8	1,060,210	1,303,478	1,574,290	1,292 bytes	2,501,500
H10 W1	232,829	237,079	240,319	8,844 bytes	293,000
H10 W2	156,705	163,563	169,667	4,620 bytes	238,000
H10 W4	185,093	204,807	223,979	2,508 bytes	363,000
H10 W8	1,065,116	1,349,696	1,579,196	1,452 bytes	2,503,000

Every observed maximum is below the budget the guard reserves, with at least 18 percent headroom for W1 and between 28 and 39 percent for the other sets. Each additional tree level adds about 300 gas and 32 bytes, so heights 15, 20 and 25 follow the same pattern. LMOTS W4 gives the best balance of gas and signature size and is the recommended default.

Committee Operations

Execution gas with LMS H10 and LMOTS W4 for committees of increasing size, each with a strict majority threshold.

Committee	Approve, hybrid	Approve, post quantum	Revoke	Rotate	Approval calldata
2 of 3	549,015	504,393	549,704	786,882	5,732 bytes
3 of 4	849,465	757,651	780,635	1,136,107	8,548 bytes
4 of 7	1,127,122	1,001,978	1,006,778	1,624,606	11,364 bytes
6 of 10	1,686,595	1,470,341	1,537,455	2,493,521	16,996 bytes

Cost grows linearly with the threshold, at about 280k gas per additional hybrid attestation and 240k gas per additional post quantum attestation. A proof of possession for a new member costs 278,975 gas and an equivocation report that freezes a misused key costs 455,679 gas.

Worst Case Rotation

A rotation must fit in one transaction under EIP 7825, which caps a transaction at 16,777,216 gas. The guard accepts a committee only when its worst case approval fits 14M gas and 120 KB of attestation data and its threshold cost plus 90k gas per member fits 16M gas. The cases below grow a committee to the largest size the guard accepts with every key new, which is the most expensive rotation possible. Figures are full transaction gas including calldata.

Parameter set	Rotation	Transaction gas	Calldata	Headroom under the cap
W4 H5	24 of 24 to 24 of 47	11,286,348	72,964 bytes	33 percent
W2 H5	25 of 25 to 25 of 49	12,435,435	128,804 bytes	26 percent
W1 H5	13 of 13 to 13 of 25	8,627,101	121,892 bytes	49 percent

The largest committee the guard accepts for each parameter set, written as threshold of members.

LMOTS	H5	H10	H15	H20	H25
W1	13 of 25	13 of 25	12 of 23	12 of 23	12 of 23
W2	25 of 49	24 of 47	23 of 45	22 of 43	22 of 43
W4	24 of 47	24 of 47	24 of 47	24 of 47	24 of 47
W8	5 of 9	5 of 9	5 of 9	5 of 9	5 of 9

Shield Delivery

Execution gas to accept one message once its batch root is approved, with a batch of 256 messages and a Merkle proof of eight nodes.

Operation	Execution gas
Inclusion check against an approved root, any bridge	7,078
LayerZero, submit with proof and forward to the receive library	128,683

The cost of post quantum verification is paid once per batch, so the cost per message falls as batches grow. With 256 messages per batch, a 2 of 3 committee adds about 2,100 gas of approval cost to each message.

Signer Kit Performance

Parameters	Key generation	Signing	Verification	Signature
H5 W1	0.07 s	81 ms	0.2 ms	8,684 bytes
H5 W2	0.06 s	79 ms	0.2 ms	4,460 bytes
H5 W4	0.07 s	82 ms	0.3 ms	2,348 bytes
H5 W8	0.19 s	88 ms	2.4 ms	1,292 bytes
H10 W1	0.38 s	82 ms	0.2 ms	8,844 bytes
H10 W2	0.40 s	85 ms	0.2 ms	4,620 bytes
H10 W4	0.68 s	83 ms	0.3 ms	2,508 bytes
H10 W8	4.72 s	90 ms	2.3 ms	1,452 bytes

Signing time is dominated by the durable writes the kit performs before it releases a signature. It locks the key, advances the high water mark and reserves the leaf on storage flushed to the device, then signs and verifies its own output. At about 80 ms per signature a committee member can attest to more than ten batches a second, far beyond the rate any bridge requires.

Live Smoke Test

The smoke test starts a local Ethereum node, generates four committee keys with the signer kit, deploys the guard and the LayerZero verifier, and runs the complete life cycle. Gas figures are full transaction gas.

Step	Result	Gas
Local node running with a funded test account	Passed	
Four signer keys generated, LMS H10 W4 with secp256k1	Passed	
Guard deployed	Passed	
LayerZero verifier deployed	Passed	
Kit leaves match the contracts for a batch of four messages	Passed	
Batch approved by 3 of 4 hybrid attestations after the minimum root age	Passed	1,009,001
Replayed attestations change nothing and cannot approve another root	Passed	
Message accepted by its inclusion proof and a forged message refused	Passed	
LayerZero packet verified and forwarded to the receive library	Passed	160,619

Step	Result	Gas
Batch revoked by quorum	Passed	955,455
New member proof of possession	Passed	343,304
Committee rotated to post quantum only mode	Passed	1,008,721
Post quantum only approval after rotation	Passed	944,387

Test Coverage

- The contract suite covers quorum counting, ordering, classical and post quantum signature failures, replay across contracts, chains, epochs and actions, leaf reuse and freezing, all or nothing actions, proof of possession, retained members, the minimum root age floor, proof depth bounds, revocation, the LayerZero verifier, the treasury controls and fuzzed Merkle proofs.
- The parity suite runs 24 randomised rounds and proves that the contracts and the signer kit produce identical root, revocation and configuration digests, LayerZero leaves, Merkle roots and proofs.
- The kit suite covers reserve leaves, crashes, in place restores, copies, clones, rebinding, retirement tombstones, a monotonic high water mark, eight concurrent signer processes, access control lists, symbolic links, network file systems, and a damaged journal.
- The RFC 8554 suites verify the published test vectors on chain and in the kit, reproduce the published public key of test case 2, and reject every altered message and signature.

Every suite runs in continuous integration on every push, and the commands to reproduce each result are listed in the repository.